A First Course in Abstract Algebra

Section 32: Geometric Constructions

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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32 Geometric Constructions

Exercise 32.1: Deriving the cosine triple-angle identity

Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta.$$

Substitute $\sin^2\theta = 1 - \cos^2\theta$ and collect terms:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta + 3\cos^3\theta = 4\cos^3\theta - 3\cos\theta.$$

The imaginary terms in the cube do not contribute to the real part.

Exercise 32.2: True and false statements

(a) True for a nondegenerate cube. If its constructible edge is $a > 0$ and the doubled cube had constructible edge b, then $b/a = \sqrt[3]{2}$ would be constructible, contrary to its degree three.

(b) True. Since even a unit cube cannot be doubled, no procedure can double every constructible cube; in fact (a) gives the stronger individual impossibility.

(c) True for a nonzero radius. Squaring a circle of constructible radius r would construct a square edge $r\sqrt{\pi}$; division by r and squaring would make π constructible and therefore algebraic, a contradiction.

(d) False. A right angle can be trisected into 30° angles, which are constructible.

(e) True. A constructible number lies in a tower of quadratic extensions, so its degree divides a power of two and is itself a power of two.

(f) False. Only a necessary degree condition was proved; being of power-of-two degree alone does not establish membership in a tower of quadratic extensions.

(g) True. The final obstruction uses that a degree divisible by three cannot divide a power of two, a consequence of prime factorization.

(h) True as the intended methodological observation: counting dimensions rules out constructions without trying all diagrams.

(i) True. Each individual constructible number has a finite construction by definition.

(j) False. A finite sequence gives only finitely many constructed points or segments, while the constructible numbers already include infinitely many integers. There is no common finite construction listing all of them.

Exercise 32.3: Why a regular nonagon is not constructible

If a regular 9-gon were constructible, its central angle 40° would be constructible. Let $u = 2\cos 40^\circ$. Since $3 \cdot 40^\circ = 120^\circ$, the triple-angle identity yields

$$u^3 - 3u = 2\cos 120^\circ = -1, \qquad u^3 - 3u + 1 = 0.$$

The monic cubic has only possible rational roots ± 1, and neither is a root. Thus it is irreducible over $\mathbb{Q}$ and u has degree three. Every constructible number has degree a power of two, so u and the regular 9-gon are not constructible.

Exercise 32.4: Constructing thirty degrees algebraically

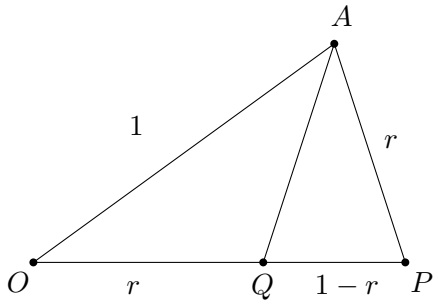
The numbers $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$ are constructible: constructible numbers include rationals and are closed under positive square roots. Hence the unit-circle point $(\sqrt{3}/2, 1/2)$ is constructible, and the ray from the origin through it makes a 30° angle with the positive horizontal axis. This gives the requested algebraic justification.

Exercise 32.5: The golden-ratio chord of a decagon

In the source triangle, $OA = OP = 1$, $\angle O = 36^\circ$, and $AP = r$. Thus the base angles at A, P are 72°. The bisector AQ gives $\angle QAP = 36^\circ$, so $\triangle OAP \sim \triangle APQ$ with correspondence $O \leftrightarrow A$, $A \leftrightarrow P$, $P \leftrightarrow Q$. Also $AQ = AP = r$, and $OQ = AQ = r$, so $PQ = 1 - r$. The similar triangles give $1/r = r/(1-r)$, hence $r^2 + r - 1 = 0$. Positivity yields

$$r = \frac{\sqrt{5} - 1}{2},$$

a constructible length. A chord of this length in a unit circle subtends 36°; the law of cosines equivalently gives $\cos 36^\circ = 1 - r^2/2 = (1 + \sqrt{5})/4$. Thus one can construct the decagon by stepping this chord around the circle. Joining alternate vertices gives a regular pentagon.



Exercise 32.6: Constructing a regular twenty-gon

Exercise 5 constructs the central angle 36° of a decagon. Bisect it to obtain $18^\circ = 360^\circ/20$, using the standard compass-and-straightedge angle-bisector construction. Repeatedly mark this angle around a circle to obtain the twenty equally spaced vertices. Thus the regular 20-gon is constructible.

Exercise 32.7: Constructing a regular thirty-gon

A regular hexagon gives 60° and a decagon gives 36°. Subtracting constructible angles gives 24°, then bisecting gives $12^\circ = 360^\circ/30$. Both subtraction and bisection are standard straightedge-and-compass constructions. Marking this central angle around a circle constructs the regular 30-gon.

Exercise 32.8: Trisecting seventy-two degrees

The angles 60° and 36° are constructible, so their difference 24° is constructible. Since $72^\circ = 3 \cdot 24^\circ$, this gives a trisector. For the given 72° angle, copy 24° from one side twice inside it. Thus some angles that are readily constructible can also be trisected, despite the impossibility of a universal trisection procedure.

Exercise 32.9: Constructing a regular fifteen-gon

Subtract the decagon angle 36° from the hexagon angle 60° to construct 24°. This is $360^\circ/15$, so fifteen equal successive central angles locate the vertices of a regular 15-gon. Equivalently alternate vertices of the constructible regular 30-gon give it.

Exercise 32.10: A brief explanation for a geometry teacher

Every coordinate obtainable by compass and straightedge is built from rational numbers using arithmetic and repeated square roots. Consequently its algebraic degree must be a power of two. If 60° could be trisected, the constructible number $u = 2\cos 20^\circ$ would satisfy $u^3 - 3u - 1 = 0$, an irreducible cubic since neither possible rational root 1 nor -1 works. Its degree would be three, contradicting the power-of-two condition.